Databases

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Database & Database Management System (DBMS)

Databases are among the most important factors in the development and advancement of information systems. Data from around the world is received in databases, and databases also feed other systems and reports for various purposes. The different programs that feed databases and the different purposes for which they exist necessitate that they take several forms and models.

1 - Data :-

Data is numbers, letters, symbols, or words that can be processed by a computer, such as the number (65).

٢- Information

Information is data that has been organized or processed to achieve maximum benefit from it.

Example: The numbers (6) and (5) are data that, when used in the multiplication operation 65, become useful information.

٣- Databases: .

One of them is the collection of a large amount of information or data and its presentation in one or more ways to facilitate use.

Another definition is a set of data and information stored in a standardized way without duplication and connected to each other according to relationships. Databases are managed by a system called a Database Management System. Data is retrieved using commands from a query language

Example: A telephone directory that includes names, addresses, and telephone numbers for a specific city can be considered a database. This database can be utilized by entering the subscriber's number to obtain their name and address, or by entering the subscriber's name to obtain their telephone number and address, and so on.

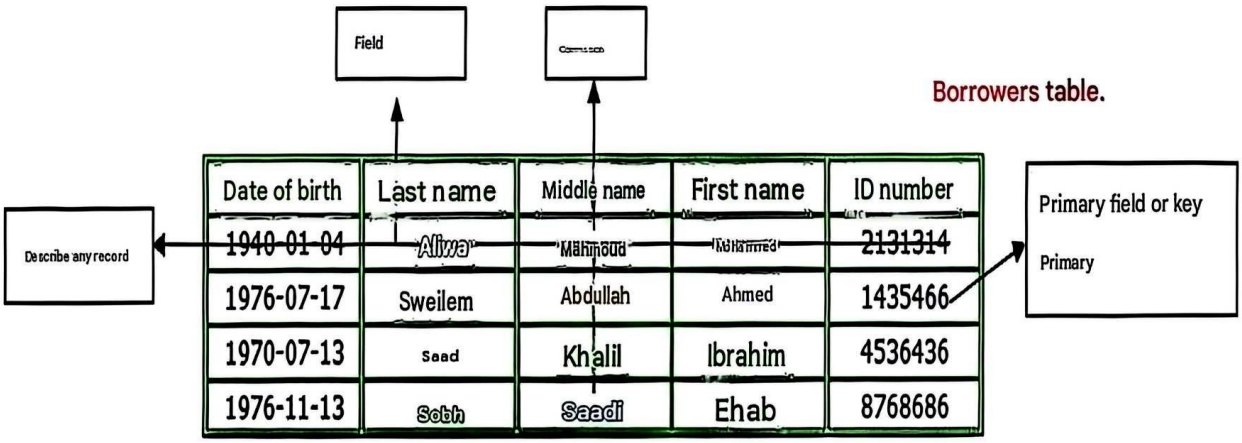
A database consists of:

١- Tables 2- Records 3- Fields

Tables: These are a collection of records (each record is a row within the table). These records contain information of the same type. A telephone directory can be considered one table.

Records: A record is a single row in the table and contains a set of fields. Any row on any page of the telephone directory can be considered a record.

Fields - (Fields) are specific elements within a record, which may be (in a phone directory, for example) a citizen's name, address, or telephone number. They are defined in terms of type and size by the program designer.



Methods of linking relationships in databases:

Every table we create must have what is called a primary key. This key is unique to each table; each user has a primary number that identifies them, and we use it to specify their name and other information. In order to link them to the related table, we make this value a foreign key. :(FOREIGN KEY)

CUSTOMERS

CustomerID	Name	Address	City
1	Julie Smith	25 Oak Street	Airport West
2	Alan Wong	1/47 Haines Avenue	Box Hill
3	Michelle Arthur	357 North Road	Yarraville

ORDERS

OrderID	CustomerID	Amount	Date
1	3	27.50	02-Apr-2000
2	1	12.99	15-Apr-2000
3	2	74.00	19-Apr-2000
4	4	6.99	01-May-2000

Note that the CustomerID column in the CUSTOMERS table is a primary key and cannot be duplicated. However, in the ORDERS table, it is a foreign key and can be duplicated because multiple orders can be made by a single user, but each order is associated with only one user.

- Importance and Advantages of Databases:

- Storing all data related to all activities of an entity in an integrated and accurate manner, classifying and organizing this data to facilitate future retrieval.

- Monitoring changes that occur in stored data and making necessary modifications to ensure it is always in the appropriate format for immediate use.

- Storing a massive amount of data that exceeds human capacity to remember its details, and then performing certain operations and processing that are impossible to execute manually.

- Helping to store data in an integrated manner, meaning linking different types of data representing all activities.

- Helping to achieve complete confidentiality of stored data so that no information is available to anyone who is not entitled to access it.

- The possibility of reducing unjustified data duplication.

- Achieving the principle of data sharing and maintaining data integrity.

- - Functions of Databases:

- Adding new information or data to the file.

- Deleting old data that is no longer needed.

- Changing existing data based on new information.

- Searching for and querying specific information.

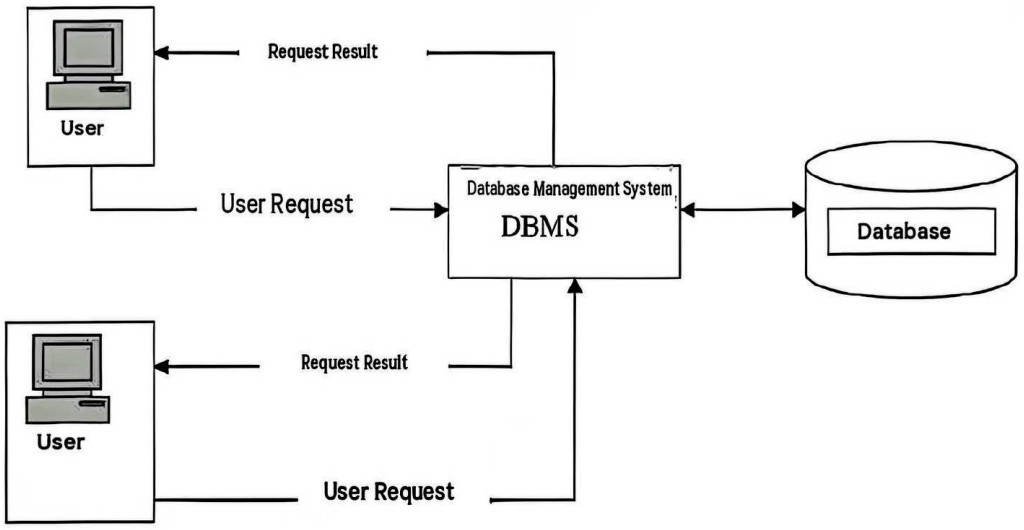
- Sorting and organizing data within files.

- Displaying data in the form of organized reports or forms.

- Calculating the final total, subtotal, or arithmetic mean of required data.

-:Database Management Systems (DBMS) -

A database management system is a set of programs that manages and controls the process of storing and retrieving data. It also provides access to and interaction with a large number of users to the database. It is considered a link between users and the database, receiving user requests, transferring them to the database, executing the necessary programs to fulfill these requirements, and then providing the user with the desired results. Examples include: Oracle, SQL Server, MS Access, FoxPro...



The difference between a database and a database management system:

A database is an electronic container where all data is stored and retrieved simply and easily.

A database management system is divided into a data definition language and a data manipulation language. The latter allows for deletion, addition, creating tables and fields, and establishing relationships between tables.

Database Management Systems (DBMS) Functions

- Creating Databases
- Updating Databases
- Running Databases
- Maintaining Databases
- Protecting and Ensuring Data Integrity
- Adding New Information or Data to a File
- Deleting Old Data That Is No Longer Needed
- Changing Existing Data Based on Newly Created Information
- Searching for and Querying Specific Information
- Sorting and Organizing Data Within Files
- Displaying Data in the Form of Reports or Structured Templates
- Calculating the Total, Subtotal, or Arithmetic Mean of Required Data

Database System Components

A database system is divided into five sections:

1 - Hardware: .

This includes all the physical devices in the system, such as computers, peripherals, printers, and communication devices in the extensive database environment, etc.

2 - Software:

This is the set of programs used in the database and is divided into three sections:

- Operating Systems: These are the programs that manage and configure the devices to work and enable the rest of the programs to run.

Linux, Unix, Windows, etc.

B - Database Software: This is the program that manages the database, such as Oracle, Sybase, DB2 ....

- Application and Utility Programs: These are the programs that perform retrieval, storage, and report generation.

٣ - Users: These are the people who work in the database environment, and they are:

- System Administrator: This is the person responsible for managing the overall environment in which the database system operates and does the following:

1. Manage users and grant permissions to use the system.
٢. Manage storage devices and other devices.
٣. Monitor system operation.

B - Database Administrator: This is the person responsible for managing the database, and their duties include:

1. Determine the database requirements in terms of software and hardware.
٢. Monitor the database system and coordinate its use.
٣. Provide security and protection for the system.
4. Design database maintenance mechanisms and define the procedures necessary to provide services to other users.

C. Database Designer: This is the person (or people) who designs the database. Their duties include:

1. Determining the data to be stored in the database.
2. Designing the best structures for storing the data.
3. Designing a redundant database.
4. Determining the methods for accessing, processing, and retrieving data by designing the screens and reports to be used.
5. Documenting the design process and methods for accessing data.

Programmers and Systems Analysts: These are the people who design and implement programs. Their duties include:

1. Designing applications and converting them into programs using a programming language(s) according to the policies established in the design process.
2. Implementing and applying these programs and ensuring their integrity.
3. Performing the necessary maintenance on these programs.

End User: This is the person or group of people who perform daily work on the system and apply the programs in a specific area, such as retrieving, modifying, deleting, generating reports, etc.

Procedures and Operations: These are the rules and instructions that govern the proper functioning of the database and are presented as clearly and specifically documented instructions.

Data: This is the most important component of the system, as it includes the set of facts stored in the database Since the data is in a primitive form, it is necessary to determine where and how to store this data to facilitate its processing and utilization. This is the designer's job.

Importance of Database Design:

Building a good database is not easy. Significant effort is required to obtain a good database. A well-designed database facilitates its use and management, while a poor design leads to data duplication (meaning the same data exists in multiple locations). This makes maintaining data consistency difficult, and data duplication usually results in incorrect results when the data is retrieved from the database. This, in turn, leads to flawed administrative decisions and future planning based on inaccurate information.

Database System Lifecycle:

1- Initial Study of the Existing System, including:

- a Analyzing the current situation of the organization and understanding the nature of the procedures, instructions, and work rules.

b- Identifying the problems facing the current system and the material constraints, such as human resources and available funding, for developing or replacing the existing system.

c- Defining the objectives to be achieved and the required features of the new system.

2- Database Design. This stage is considered one of the most important in the system lifecycle. Significant effort is required to design the system effectively and achieve its intended objectives. The design process includes the following:

A - Building the Concept Model. This process includes several steps (we will discuss this process in detail in later chapters):

Analyzing data, user requirements, and required procedures

Defining and identifying entities, their characteristics, and their relationships to each other, as well as placing them in a standard format.

Drawing the concept diagram, which is a graphical model that describes the system entities and their relationships to each other.

٤. Modifying the model to include the main procedures, rules for adding, modifying, and deleting data, reports, and screens, and the degree of data sharing and compatibility...

B - Selecting a Database Management System (DBMS).

c) Converting the conceptual model into an internal model using a Database Management System (DBMS). -

٥ - Physical design, which involves defining storage specifications, storage media, and data access methods using the DBMS.

٣ - System implementation: This stage involves creating tables and writing all the necessary programs to implement the system's requirements, including different screens and required reports.

٤ - System testing and evaluation, which includes:

a) Checking the database and ensuring it functions correctly. -

b) Evaluating the performance of the programs and applications used.

e) System deployment in the workplace: This process includes creating tables, users, and permissions, and installing and implementing all programs and applications in the actual environment where the system should operate.

٦ - System monitoring: This process continues throughout the system's lifecycle to ensure it functions correctly and to modify the system to meet new workplace requirements, such as changes in laws, regulations, and work rules.

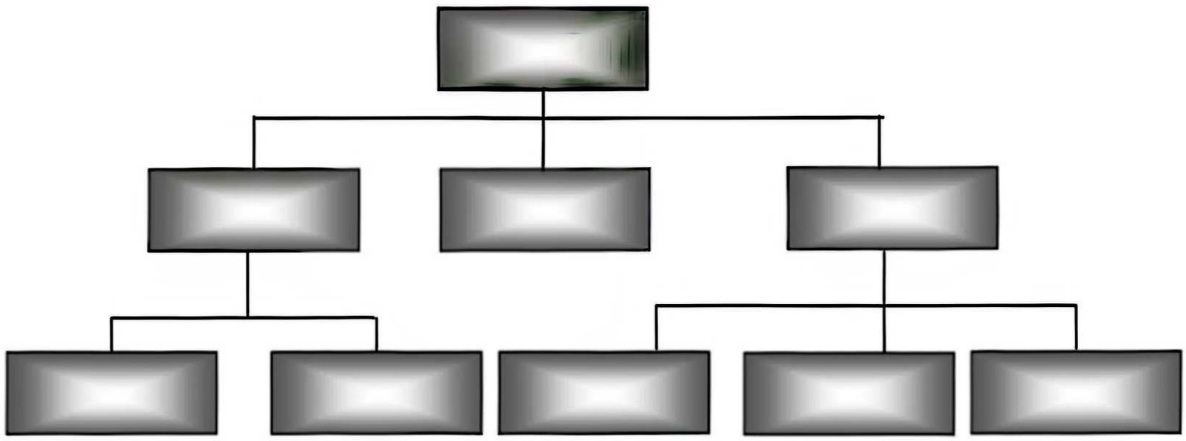
Types of Databases:

1. Manual Databases

The scattered data on shelves and papers that fill libraries and warehouses represents manual databases. Because of this dispersion, important and logical classifications have emerged to categorize data according to their relationships to each other or their mathematical or logical structure. Based on what I mentioned, files stored on shelves, electronically scheduled sales invoices, and people's addresses in a document file are examples of manual data.

1. Hierarchical Databases

This database relies on the principle of hierarchical operation. It creates a sequence from the origin or root, where this system branches out into sections. Its operating principle is based on accessing data either from the bottom up or from the top down, in a sequential and branching manner.



Hierarchical Database

Disadvantages of the Hierarchical Model:

1- It lacks flexibility and good user responsiveness.

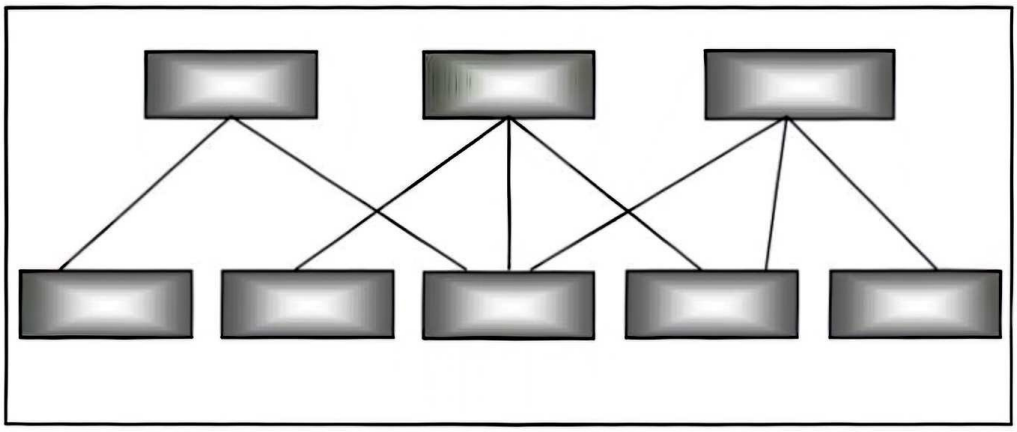
. 2- Programming complexity

3- Data is stored in a hierarchical structure, making it difficult to change or modify this structure.

4- It does not provide adequate support for unstructured queries.

Network Databases ٣

Data is stored in the network structure as interconnected chains of data, thus representing more complex logical relationships. Network databases are still used with database management systems for large computer systems. This database represents a pattern of many-to-many relationships between records.



Network Database Model

Disadvantages of the Network Model: Its disadvantages include inflexibility and difficulty or complexity in terms of programming and maintenance. However, it processes information efficiently.

4. Relational Databases:

The concept of relational databases began to emerge in 1970 when Codd proposed this model, which was built on the theories of relational algebra. This is where the strength of this model and its rapid spread became apparent. In the early 1980s, many companies began to adopt and implement this model, and we now observe that most database systems on the market conform to it. The idea behind the model is to view the database as a collection of tables or relationships called "relationships" (hence the name of the model). Each table must have a name; no more than one table can have the same name. A relationship is a mathematical term and represents a table with two dimensions (rows and columns). The order of the rows or columns is not important Rows represent the records or tuple of a table, and columns represent the attributes of that table. Each attribute must have a domain of values that the column can contain. These tables are linked together by links. Each table must have a primary key to distinguish the rows from each other. The point that represents the intersection of a row and a column (attribute) represents a value for that row. In the rest of this unit, we will provide a description of relational databases in terms of their components and their most important characteristics.

The following table represents student information in a university database.

Table Name: Student -

Each row represents information for only one student. -

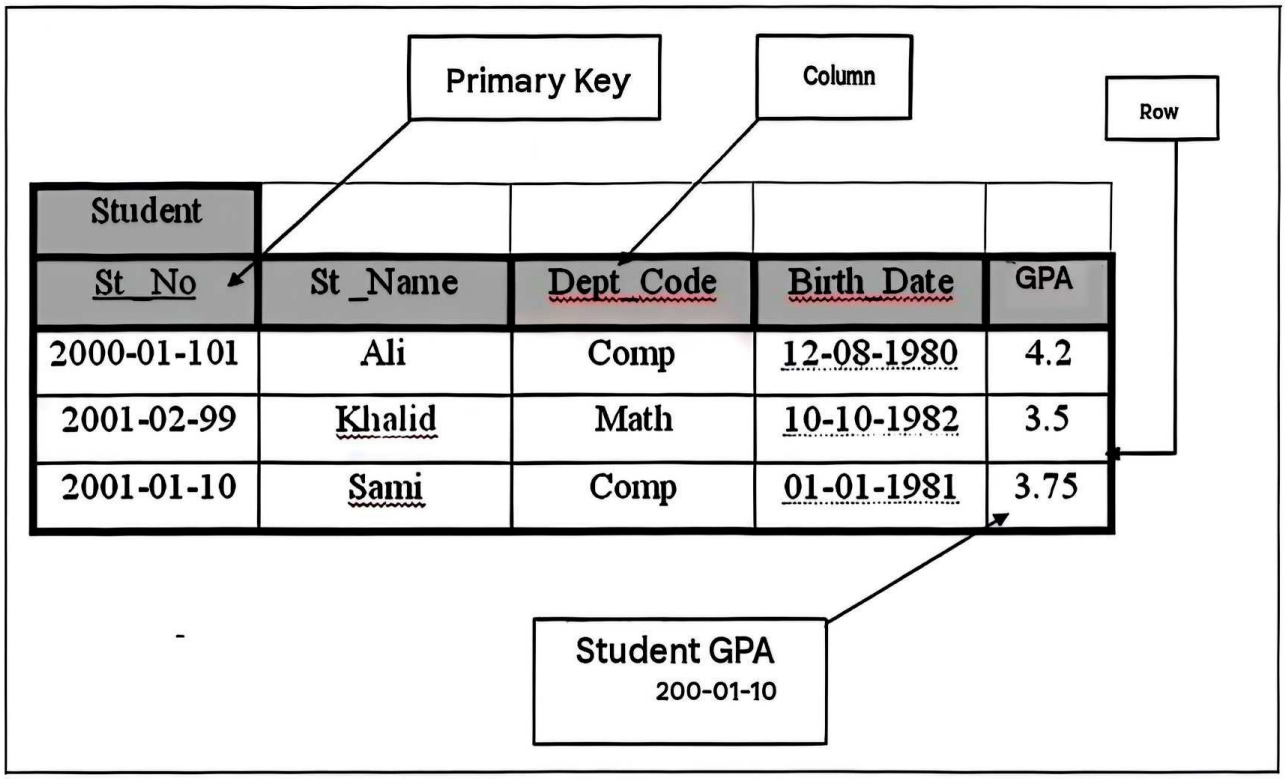
The primary key for the table is St_No. Each student must have a different number from the other students. -

. The attribute Dept_Code represents the department to which any student belongs. -

- The point where the attribute (GPA) of the column intersects with the third row represents the cumulative grade point average of student number 2001. -

01-10

Domain of Values: Each attribute must have a fixed domain of values. For example, GPA must contain - a real number between... 5. The Dept_Code department must be one of the academic departments at the university.



- The order of the rows or columns is not important. For example, the previous table could be on

The following figure:

Student				
St_No	St_Name	GPA	Birth_Date	Dept_Code
-01-2001 10	Sami	3.75	1981-01-01	Comp
-02-2001 99	Khalid	3.5	1982-10-10	Math
-01-2000 101	Ali	4.2	1980-08-12	Comp

Concepts of Relational Databases:

- Tables consist of records and fields and are related to each other in several ways, hence the name relational databases.
- A record is a row that contains all the information about an element of the database. For example, in a student table, each student will have a record that identifies them.
- A field is a column in a table and represents an attribute of a student, such as their ID number, name, or department at the university.
- A relationship is a way of linking tables together to collect information.

Characteristics of Relational Databases:

- Each table in a relational database contains a repeating type of record.
- There is no specific order for the fields.
- There is no specific order for the records.
- Each record has one or more fields.
- If the data in one table changes, the data in all related tables will change.

The difference between hierarchical, network, and relational databases:

Both hierarchical and network models use links or pointers to connect records within the system. These systems are called static or monolithic because the records are physically linked through their definitions.

These systems are characterized by their complexity and difficulty in modification, but their speed of access compensates for these drawbacks.

In relational systems, however, the linking of records is not done physically through pointers, but rather through the actual names of the fields, such as employee number, name, or card number. Records in this system are connect-addressable, meaning they are accessed by matching the values of the stored data with each other.

Table Keys (Relationships):

Keys are among the most important features of relational databases, as they are the distinguishing feature of a specific table on the one hand, and the link that connects different tables to each other on the other. Keys in relational databases can be divided into several categories:

- Super Key: This is the smallest set of attributes that can distinguish a row in the table from all other rows. For example, this set of attributes can be a super key. A

St_No
St_No, St _Name
St_No, dept_code

B - Candidate Key: This is an attribute (a set of attributes) that can be chosen as the primary key for the table. There must be more than one row with the same value for this attribute or attributes, and it must also have a value (not null).

However, as we have seen, St_No, St_Name is a super key, but it is not a candidate key to be a primary key because St_No alone is sufficient to distinguish any row from all other rows. Therefore, St No is considered a candidate key to be a primary key

- Primary Key: This is the key chosen from a set of candidate keys to identify each row in the table. We can choose St No. as the primary key. C.

- Secondary Key: This is an attribute or attributes used for retrieval purposes. For example, if we have a table containing a list of customers, the primary key is the customer ID (Customer_id). D

But if we want to retrieve the phone number of a specific customer (but who will remember customer numbers?!), in this case, the name is usually used in the search process, not the number. So, the customer's name is chosen as as the secondary key.

Customer id	Customer name	tel	Address

- Foreign Key: This is an attribute or attributes that refer to a primary key or a unique value in another table. For example, the attribute (Dept Code) in the Student table is a foreign key for the Department table. E.

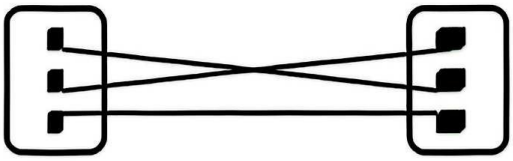
Student				
St _No	St _Name	GPA	Birth_Date	Dept_Code
2001-01-10	Sami	3.75	01-01-1981	Comp
2001-02-99	Khalid	3.5	10-10-1982	Math
2000-01-101	Ali	4.2	12-08-1980	Comp

Department	
Dept_Code	Dept_name
Comp	Computer
Math	Mathematics

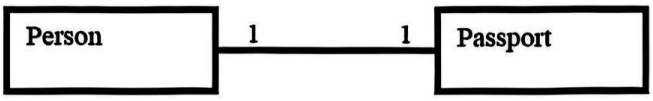
Interrelationships between tables (relationships):

This represents the degree to which the tables are related to each other. These relationships must clearly define how these tables are related to each other. There are three degrees of table relationship:

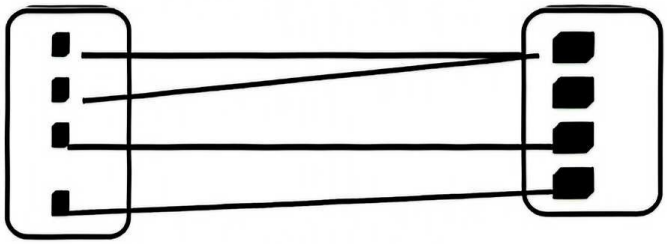
1:1: This means that one value in the first table corresponds to only one value in the second table



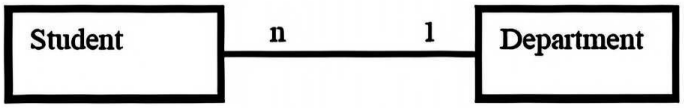
For example, we can specify that each person has only one passport and that the passport belongs to only one person .



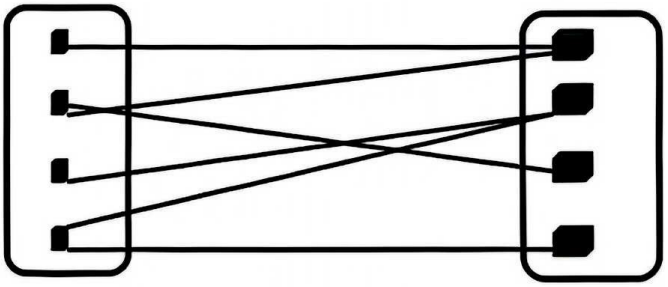
٢. One-Many or Many-One (1:N or N:N): This means that a value in the first table corresponds to a value in the second table, and that a value in the second table can correspond to one or more values in the first table.



For example, a trainee must belong to only one department, but at the same time, there can be more than one student belonging to that department.



٣. Many-Many (N:N): This means that a value in the first table corresponds to one or more values in the second table, and that a value in the second table can correspond to one or more values in the first table.



For example, a student can register in more than one section, and more than one student can register in a section.

